



DIGITAL TECHNOLOGY

Academic Year 2024-25

Prof. Barbara PERNICI

Exam 3/9/2025

Total time 1.30 h

Last name

First name

Matricola/Person code

Signature

Please remember that:

- *The exam is closed books.*
- *The use of cellular phones or any other electronic devices during the exam is forbidden.*

It is necessary to answer at least partially all the questions for a positive evaluation.

Given answers should be explained, lists of bullet items are insufficient to answer a question.

Please write the answer for each question on a separate piece of paper.

Question 1 [11 points]

Discuss data quality problems in time series, giving examples and explaining the challenges posed by each problem.

Question 2 [11 points]

Illustrate the value of data/information according to the Moody's laws studied in the class.

[please turn over]

Question 3 [11 points]

Consider the following fragment of Python code:

```
import pandas as pd

names = ["Einstein", "Turing", "Lovelace", "Newton", "Curie"]
scores = [14, 28, 19, 30, 17]

def evaluate(score):
    if score >= 18:
        return "Pass"
    else:
        return "Fail"

results = []
for s in scores:
    results.append(evaluate(s))
```

Questions:

- 1) Explain how the *evaluate* function works, and the content of the *results* variable after executing the code.
- 2) Modify the function so that it returns "Excellent" if the score is greater than or equal to 27.
- 3) Using Python, construct a Pandas DataFrame with column names "*Name*", "*Score*", and "*Result*" and count how many records fall into each "*Result*" category. Show the result.